

Brain-Machine Interface for neurorehabilitation and human augmentation

Applications of BMI technology and prospects

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ABSTRACT

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KEYWORDS

Brain-Machine Interface, Brain-Computer Interface, EEG, neurorehabilitation

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1 ABSTRACT

Brain-Machine Interface (BMI) explores the potential of recording and analyzing brain waves in real-time using AI to visualize and manipulate neural activity for motor recovery in stroke patients. The methodology involves placing sensors on the scalp to measure brain waves, which are then processed through AI algorithms to extract meaningful brain activity patterns. By identifying undamaged residual neural circuits, the robot attached to the paralyzed fingers provides movement assistance, along with somatosensory feedback through electrical stimulation. This approach aims to stimulate the

creation of new neural connections or rewiring in the brain, leading to improved motor function and recovery.

Traditional medical approaches for severe motor paralysis caused by stroke often face challenges in achieving significant recovery. However, the use of AI-enabled brain wave analysis provides a promising solution. By visualizing and understanding the brain's neural activity, clinicians and therapists may be inspired to design personalized treatment programs that target specific neural circuits associated with motor function.

Moreover, the application of this technology extends beyond medical settings. It has also shown potential in the field of human augmentation, with efforts focused on enhancing sports performance and musical skills. By leveraging the ability of the brain to adapt and rewire itself, athletes can potentially improve their coordination, precision, and timing. Similarly, musicians may enhance their own skills, leading to improved dexterity and musical expression.

While the practical implementation of this technology is still under development, the results from initial studies and trials are encouraging. Continued research and refinement of AI algorithms, as well as advancements in neuroimaging techniques, will likely contribute to the future practical application and widespread adoption of this technology. Ultimately, it holds the potential to significantly improve the quality of life for patients with neurological disorders and enhance human capabilities in various domains.

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